Behavioral validation of generative corporate distress scores: A seven-dimensional protocol

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Highlights

- Rewriting the same figures changes which firms a generative score flags.
- The generative score ranks distress risk but trails conventional models.
- Predictive accuracy and usefulness for screening are different properties.
- A seven-dimensional protocol links tool behavior to screening decisions.

Abstract

Screening firms for financial distress uses scarce analyst time, so a risk score should give the same decision whenever a firm's facts are the same. We ask whether a score from a generative artificial intelligence (AI) model does. Using accounting and market data on Chinese listed firms and their later distress outcomes, we compare a generative AI scoring tool with conventional models and test it by writing each firm's figures in equivalent ways, adding unverified text, and changing technical settings. The generative score identifies firms that later become distressed, but less well than the conventional models. Under one decision rule, the firms flagged, the review workload, and the share of later distress events captured all depend on how the same figures are written. We propose a seven-dimensional validation protocol that links the tool's behavior to screening decisions.

Keywords: corporate distress; generative artificial intelligence; model risk; model validation; probability calibration; representation stability

JEL Classification: C52; G21; G32; G33

1. Introduction

Lenders, investors, and supervisors use risk scores to decide which firms receive review and financing. A conventional model applies a fixed formula to a firm's financial variables, so the same numbers always give the same score. A generative artificial intelligence (AI) model reads text instead, and the same figures can be written in many ways, in a different order, with a ratio stated as 12.5% or as 0.125, or as a list, a table, or sentences. We call each such way of writing

the same figures a representation. A financial ratio is hard information (Liberti and Petersen, 2019), but once a generative model reads it as text, its score can depend on wording that says nothing about the firm.

One firm from our sample shows the problem. With its ten variables listed in our baseline order, the model returns a score of 32, which our calibration converts into a distress probability of about 1.26%. With the same lines in reverse order, the score falls to 12, about 0.52%, and with ratios written as decimals, it is 38, about 1.52%. An institution that reviews every firm with a probability of at least 1% would review this firm under two of the three representations and pass over it under the third, although nothing about the firm has changed.

We develop a validation protocol, a set of tests that an institution runs before it relies on a generative AI score for a stated use. The object of validation is one fully specified configuration, the model together with every setting that can affect its reply. We apply the protocol to Chinese listed firms, where exchanges place a firm under special treatment, marked ST or *ST, when its financial condition, audit opinion, or conduct breaches the listing rules. A generative AI model and three conventional models receive the same ten accounting and market variables and score the risk of a first designation in the following year. The generative score separates later designated firms from others, but less well than the conventional models. When the same figures are listed in reverse order or written as decimals, the score changes for most firms, and a fixed threshold or a fixed number of reviews selects different firms and captures a different share of later designations. These decision results come from our primary configuration, Qwen3-4B, and three further configurations also change raw scores across representations.

The problem is economic. Because monitoring is costly, institutions decide which firms to review by applying a threshold to a score, and where a borrower falls relative to a threshold can matter (Keys et al., 2010). The value of a forecast lies in the decisions it supports (Elliott and Timmermann, 2008; Kleinberg et al., 2018; Stein, 2005), so a score that ranks outcomes well can still serve screening poorly if equivalent representations of one firm fall on different sides of a threshold. We ask whether generative scoring makes the score depend on presentation in this way.

Distress prediction research now uses dated risk sets, time-varying covariates, and out-of-sample evaluation (Bharath and Shumway, 2008; Campbell et al., 2008; Duffie et al., 2007; Kim et al., 2022). Machine learning improves ranking (Berg et al., 2020; Kelly et al., 2024; Khandani et al., 2010; Lessmann et al., 2015) and changes who receives credit (Bartlett et al., 2022; Fuster et al., 2022), and recent work extends these tools to forecasting, interpretation, Chinese distress, scenarios, and global training data (Börner et al., 2026; Cakici and Zaremba, 2026; Chun et al., 2025; Kang et al., 2026; Kim et al., 2025; Kuang et al., 2026). Language models extract information from disclosures, news, and speeches (Blankespoor et al., 2026; Cao et al., 2024; Fieberg et al., 2026; Lopez-Lira and *Tang*, 2026; Siano, 2025) and change how investors trade (Cheng et al., 2025). The closest study, Cruz et al. (2024), finds that language-model risk scores carry signal but give miscalibrated probabilities that depend on how the question is posed, and prompt research documents similar sensitivity to formatting and wording (Jiang et al., 2021; Lin et al., 2022; Lu et al., 2022; Sclar et al., 2024; Tian et al., 2023). These studies ask what a model predicts. We instead hold one firm's facts fixed, change only their written form, and follow the change through a decision rule to the review queue, in the spirit of Bao et al. (2020), who evaluate a fraud model by the cases regulators can investigate. We also

test whether calibration, which a probability forecast requires (Gneiting and Raftery, 2007), removes the effect, and it does not.

Model-risk research treats model error and disagreement as risks that matter for decisions (Danielsson et al., 2016; Glasserman and Xu, 2014), and supervisory design, changes in lending, and technical estimation choices can all alter model outcomes (Lee et al., 2025; Lee et al., 2026; Leitner and Williams, 2023; Rajan et al., 2015). Supervisors require separate evidence on ranking, probabilities, data, stability, use, and change control (Basel Committee on Banking Supervision, 2005; Basel Committee on Banking Supervision, 2023; Prudential Regulation Authority, 2023), and revised U.S. guidance leaves generative AI to institutions' broader risk management (Board of Governors of the Federal Reserve System, 2026). None of these principles says how to test equivalent writings of the same figures, unsupported text, or technical settings.

Our contribution is a framework that tells an institution what to check, beyond predictive accuracy, before approving a generative AI score for a stated use. It measures how a tool treats the same firm under equivalent inputs, ties instability to a dated financial outcome, separates the roles of representation, calibration, technical settings, and the decision rule, and treats each test as separate evidence so that a good result cannot offset a failure. Section 2 sets out the tests, Section 3 the data, Section 4 the results, and Section 5 their risk-management use.

2. From Risk Scores to Approval Evidence

The protocol produces a profile of evidence on seven dimensions, not an index, because an average would let a strong result on one dimension offset a failure on another, such as good ranking offsetting a badly defined outcome. Such trade-offs have no common unit, and the failures are repaired by different parts of the institution, the outcome by the business and data units, unreliable output by the technical settings, reactions to unsupported text by rules on admissible text, and poor probabilities or decisions by recalibration, a revised threshold, or a narrower use. The tool is approved for a use only if it passes every one of the seven tests against a tolerance set in advance,

, (1)

where equals one if configuration is approved for intended use and zero otherwise, indexes the seven tests, is the product over the tests, which equals one only if every test is passed, is the indicator function, is the shortfall measured in test , for example the share of firms whose review decision changes when their figures are rewritten, and is the largest shortfall that the institution has agreed in advance to tolerate in that test for that use. The empirical application reports and leaves the tolerances to the institution. Because the tolerance depends on the use, the same configuration can pass for one use and fail for another. Human review does not make instability harmless. An analyst sees only the firms that a score sends to the queue, so a risky firm with a low score under one representation never reaches review, and an automatic adverse action has no check at all. The tolerance therefore depends on the cost of a missed case, on how well review corrects errors, and on whether the score acts on the firm directly. Our evidence measures the shortfalls but does not identify the optimal tolerance for any use.

The seven dimensions fall into four groups. The first concerns the outcome. Target validity

asks whether the event being predicted is the one the institution cares about and is defined without ambiguity. The second is technical. Technical reproducibility asks whether the same input returns the same output, and representation stability asks whether the score is unchanged when the same figures are written in an equivalent way. The third is behavioral. Economic response asks whether the score moves in the expected direction when a firm's finances change, and information isolation asks whether the score ignores text that contains no verified fact. The fourth concerns statistics and decisions. Probability quality asks whether the scores can be read as accurate probabilities, and action stability asks whether the review decision is the same across equivalent representations. A fixed configuration produces

, (2)

where indexes firms, indexes fixed configurations, is the verified structured information for firm , is a representation of those facts, is additional text, is the mapping from the text given to the model to the number it returns, and is its risk score on the zero-to-one-hundred scale. The configuration index covers the model, its compression (quantization), the instructions, the software that builds the input and reads the score, and the hardware. A score is not assumed to be a probability, and a reviewer must test that interpretation against a defined outcome and population, in the same way that a proposed proxy must be shown to capture the construct it is intended to represent (Lee and Ryu, 2024). Target validity specifies the event , the population of firms at risk , the date on which firms are scored, the forecast horizon, and the information that may be used. When rules or market conditions change, the outcome must be analyzed period by period because the meaning or frequency of can shift. Technical reproducibility requires that every reply yields a valid score and that identical inputs return identical outputs, within a session and after a restart. Representation stability is assessed by varying ways of writing the inputs that are declared economically equivalent before testing. In the main test set,

, (3)

where is the core set of three representations, namely the baseline representation, which lists the ten variables one per line in a fixed order with ratios in percent, the same list in reverse order, and the baseline order with ratios written as decimals. For each firm, the raw score range is

, (4)

where is the within-firm score range, indexes a representation in , is firm 's score under representation with no added text, and and select the largest and smallest scores across the core representations. Because a range grows with the number of representations, the extended test also reports pairwise differences. A local test of economic response changes one input by a small amount while everything else in the text given to the model is held fixed. The response has the wrong direction when higher leverage (*Lev*), lower return on assets (*ROA*), or lower cash holdings (*Cash*) reduces the risk score, or when the reverse improvement raises it. Sales growth (*Growth*) has no required direction because high growth can carry financing and execution risk.

The information isolation test, which we also call the unsupported-text experiment, adds a sentence that the instructions label as unverified and as containing no new fact. We vary its tone (favorable, neutral, or adverse), its wording (three versions for each tone), and its placement (before the financial facts, after them, or inside the general instructions to the model, known as the system message). Comparisons are made within each firm, so the effect of tone is separated from differences in risk across firms and from the effect of a particular wording or placement. The fourth group distinguishes four claims,

, (5)

where the signs separate four claims that must be established one by one and do not denote numerical inequalities. These are ranking, calibrated probability levels, skill over a simple reference forecast, and stable decisions under a rule set outside the model. The first three constitute probability quality. We measure ranking with the area under the receiver operating characteristic curve (AUC), the probability that a random distressed firm scores above a random non-distressed firm, and with average precision (AP). We measure the accuracy of probabilities with the Brier score, the mean squared difference between the forecast probability and the outcome coded as one or zero, and with log loss, which penalizes confident forecasts that prove wrong more heavily. Lower values are better for both. A forecast that gives every firm the sample event rate has a Brier score of , and Brier skill is

, (6)

where is Brier skill, is the model's weighted mean squared probability error, and is the corresponding error of the constant forecast. A positive value means that the model's probabilities are more accurate than the constant forecast.

Two further steps translate raw scores into probabilities and then into decisions. We first treat as a naive candidate probability and then correct it with Platt scaling. We regress the observed outcome on the log odds of that candidate, logit(s/100), in a weighted logistic regression and use the fitted values as calibrated probabilities. We refer to the fitted function as the Platt map

, (7)

where is firm 's raw score, is the calibrated probability map, is the log-odds transform, is the calibration intercept, and is the calibration slope. Candidate probabilities are limited to [0.000001, 0.999999] so that the transform is defined. The map is always estimated on one part of the sample and applied to another. Records are split into five firm-grouped folds, each calibrated on the other four. For representation tests, each fold estimates on baseline scores and applies that same map to reversed-order and decimal-unit scores, because a separate map for each representation would hide part of the instability under review. A probability threshold set by the institution defines

, (8)

where equals one if firm is flagged for review under representation and zero otherwise, is the

probability threshold, is the calibration map estimated on baseline scores, is the raw score, and is the indicator function. The crossing rate is the share of firms for which is not the same under all representations in . We report it for thresholds from 0.25% to 5%. A fixed review capacity gives a second decision rule that needs no probabilities. The institution reviews the firms with the highest scores until its capacity is used. For each representation , the review list contains the firms with the highest scores, with ties resolved by a firm identifier unrelated to the representation. We compare the three lists at 5% and 10% capacity by their intersection, their union, the Jaccard overlap (the ratio of the two), and the number of later distress events each list captures.

The framework works in four steps. The unit that plans to use the score states the use, such as a ranking that sends the top 5% of firms to review or a probability that triggers a refusal of credit. It sets in advance the largest shortfall it will accept on each dimension, given its review capacity and the costs of a missed warning and an unnecessary review. A validator who did not build the tool runs the seven tests on the fixed configuration. The configuration is approved for that use only if every result lies within its tolerance, and otherwise it is restricted, repaired, or rejected. Which tests bind depends on the use. A fixed-capacity list needs good ranking and stable lists but not accurate probability levels, whereas a probability threshold, and above all an automatic decision, also needs calibrated probabilities and stable flags. The framework is a proposed governance structure, not an optimal rule, and the affected tests are rerun after any change to the configuration and during use as event rates, inputs, and rules change.

3. Data and Design

We begin with the dates that define the sample of firms at risk and the outcome. The source data combine licensed annual financial data, annual-report announcement dates, daily stock returns, industry codes, and a file that records the date of every change in a firm's special-treatment status. The analysis uses fiscal years 2011-2024. Fiscal year 2025 cannot enter because the full one-year outcome window for its annual reports is not yet observed (Appendix A.1). For fiscal year , the scoring date is the annual-report announcement date plus 30 calendar days. A firm enters the risk set, the sample of firms that can still receive a first designation, only if its status is normal on that date. The main event equals one when the firm's first change of status during the one-year period following the scoring date is from normal to ST or *ST. A broader outcome that also counts trading-related warning and delisting is reported in Appendix Table A2.

The 30-day waiting period addresses the timing of warnings, which exchanges often apply soon after annual disclosure. If firms were scored on the announcement date, the model could simply reproduce a designation already determined by the same report. Waiting 30 days and requiring normal status excludes those immediate designations from the risk set, while the score still uses only the variables available by the announcement date. ST marks "other risk warning," while *ST marks delisting-risk warning. The 2024 rule revision introduced new triggers and modified existing ones, including warnings for some firms sanctioned for financial falsification without meeting the threshold for major-violation delisting (China Securities Regulatory Commission, 2024). We flag fiscal years 2023-2024 as rule-exposed because their one-year windows overlap the new rules.

We exclude finance-sector firms (industry code J). The panel contains 46,058 records from

5,215 firms. The 2011-2018 training sample has 20,668 records and 346 events, and the 2019-2024 holdout sample has 25,390 records and 474 events, an event rate of 1.87%. The structured models are estimated on the training sample and evaluated on the holdout sample, and the generative model is used as released, without training on our data. Calibration maps are estimated within the holdout sample by firm-grouped cross-fitting, so each firm's probability comes from a map estimated on other firms. Because the generative model's pretraining data are not public, it may have seen information about these firms, which is one reason the protocol tests behavior rather than history. Company violation and punishment records through July 2025 and annual-report inquiry letters through August 2025 provide an external check on the outcome. They count only when announced by the scoring date and are not given to any model.

Table 1. One-year holdout risk set

	Records	Events	Rate	Regime
2019	3,382	71	2.10	pre-rule
2020	3,723	46	1.24	pre-rule
2021	4,214	57	1.35	pre-rule
2022	4,554	72	1.58	pre-rule
2023	4,751	109	2.29	rule-exposed
2024	4,766	119	2.50	rule-exposed
Total	25,390	474	1.87	—

Notes. Each record is a nonfinancial firm in normal status 30 calendar days after its annual-report announcement. The event is the first transition to ST or *ST during the one-year period following the scoring date. Fiscal years 2023-2024 are reported separately as rule-exposed because the 2024 listing-rule changes can affect the outcome definition. Rows index the fiscal year of the financial record. *Records* is the number of at-risk firm-year records. *Events* is the number of first ST or *ST transitions. *Rate* is the event rate in percent. *Regime* indicates whether the one-year window precedes or overlaps the 2024 listing-rule changes.

All models receive the same ten variables, *Size* (log total assets), *Lev* (liabilities/assets), *ROA* (return on assets), *Growth* (sales growth), *Cash* (cash/assets), *Tang* (tangible assets/assets), *Q* (Tobin's *Q*), *Age* (log years since listing), *Loss* (loss indicator), and *Vol* (the within-fiscal-year standard deviation of daily returns). Continuous inputs are winsorized at the training-period 1st and 99th percentiles, and all models use the same rounded values shown in the prompt. The structured benchmarks, which take the variables directly as numbers, are a regularized logistic regression and two tree-based models, histogram gradient boosting and XGBoost (Chen and Guestrin, 2016; Pedregosa et al., 2011). Their settings are tuned on 2011-2016 against 2017-2018, and the chosen models are re-estimated on 2011-2018 and evaluated on 2019-2024.

Running a generative model on every record is slow, so we score samples of the holdout firms. The paired audit contains 256 distinct holdout firms, 64 events and 192 non-events, spread across fiscal years with one record per firm, and events are deliberately over-represented. Each firm is scored under the three core representations, the baseline list, the same ten lines in reverse order, and the baseline order with percentages written as four-place decimals. A nested sample of 64 of those firms, with 16 events and 48 non-events, receives the extended tests. These are five random orders of the variables, the figures written as structured data (JSON), as a table, and as full sentences, percentages in basis points, ten changes to the financial inputs, and the unsupported-text experiment (Appendix C lists all variants). The probability audit is designed to represent the full 2019-2024 holdout sample. It includes all 474

events and 1,026 non-events sampled within 504 cells defined by fiscal year, sector, and size quintile. Each sampled non-event receives weight , the number of non-events in its cell divided by the number sampled from it, and events have weight one. The weights sum to 25,390 and recover 474 events exactly, so weighted results, which we call design-weighted, describe the full holdout sample. Uncertainty is assessed with a bootstrap of 1,000 draws that resamples whole firms, and calibration is refit within each draw.

The primary configuration is Qwen3-4B-Instruct-2507, an openly available model with about four billion parameters, compressed to four bits per parameter (Qwen Team, 2025). The comparison of configurations adds the same model without compression (bfloat16), a larger model of the same family (Qwen3-8B), and a model from another developer (Mistral-NeMo-12B), both in four-bit form. All run locally at temperature zero, which switches off randomness in generation. The paired and probability audits submit firms eight at a time and the configuration comparison one at a time, and because batching can slightly alter the arithmetic inside the model, batch size is a setting to be tested. The instructions in Appendix C tell the model to use only the supplied variables, to ignore evaluative text without data support, and to reply with a single 0-100 score for the risk of a first ST or *ST designation in the following year. All 9,300 runs return a valid score, and Appendix E records model identifiers, software versions, and run counts.

4. Results

Table 2 reports ranking and probability quality in the design-weighted probability audit. The Qwen score from the baseline representation ranks later distress better than chance, but all three structured benchmarks have a higher AUC, and Qwen's AUC lies below each of theirs in every bootstrap draw (Appendix Table A12). The Qwen score is always a whole number and takes only a few distinct values. Read directly as a probability, the score divided by one hundred implies a mean probability many times the observed event rate, so a user must calibrate the score before using it as a probability. After Platt scaling, Qwen's mean probability is close to the event rate, and its Brier skill is positive and close to that of the structured benchmarks. A stricter check that calibrates only on earlier years whose outcomes are fully observed gives a bootstrap interval for Qwen's Brier skill that includes zero, so this check does not show a significant improvement over a forecast that gives every firm the historical event rate.

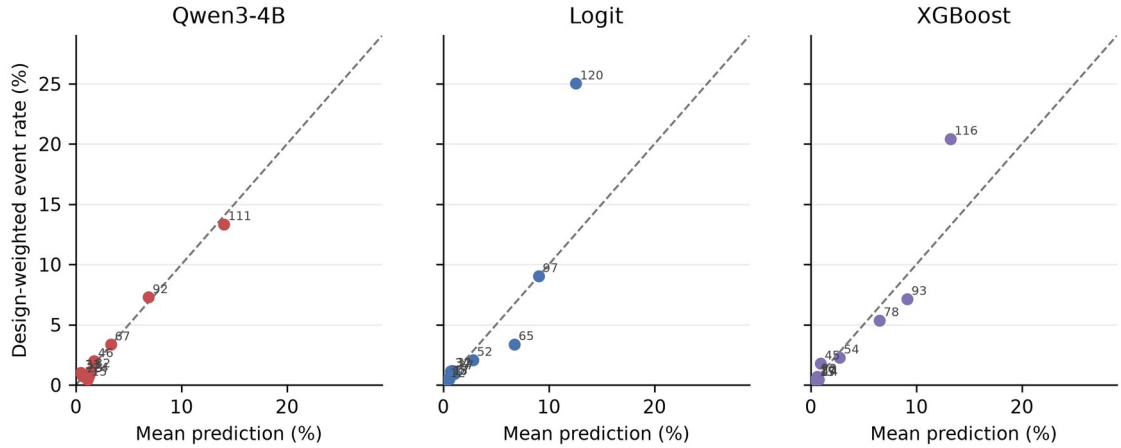
Table 2. Design-weighted ranking and probability quality

	AUC	AP	Mean	Brier	BSS	Loss
Qwen3-4B score	0.748	0.100	1.89	1.74	0.051	0.082
Regularized logit	0.810	0.144	1.88	1.74	0.053	0.079
Histogram gradient boosting	0.805	0.089	1.87	1.76	0.040	0.080
XGBoost	0.815	0.134	1.87	1.73	0.053	0.079

Notes. sampled records, including all 474 events. Sampling weights recover 25,390 holdout records and a 1.87% event rate. Ranking metrics use raw scores. Probability columns use Platt calibration estimated in five folds, with all records of a firm in the same fold. Each Qwen representation receives the map estimated on baseline scores. No separate map is estimated for any representation. *AUC* is the raw ROC-AUC and *AP* is average precision, both computed on raw scores. *Mean* is the mean calibrated probability in percent. Brier is the Brier score multiplied by 100. *BSS* is the Brier skill score. *Loss* is the log loss.

Figure 1 compares forecast probabilities with observed event rates after Platt calibration for ten groups of firms sorted by forecast probability, and points on the diagonal indicate accurate probabilities. Calibration corrects the large error in the average level, and the groups with the highest forecasts do contain more events, but no model lies on the diagonal in every group.

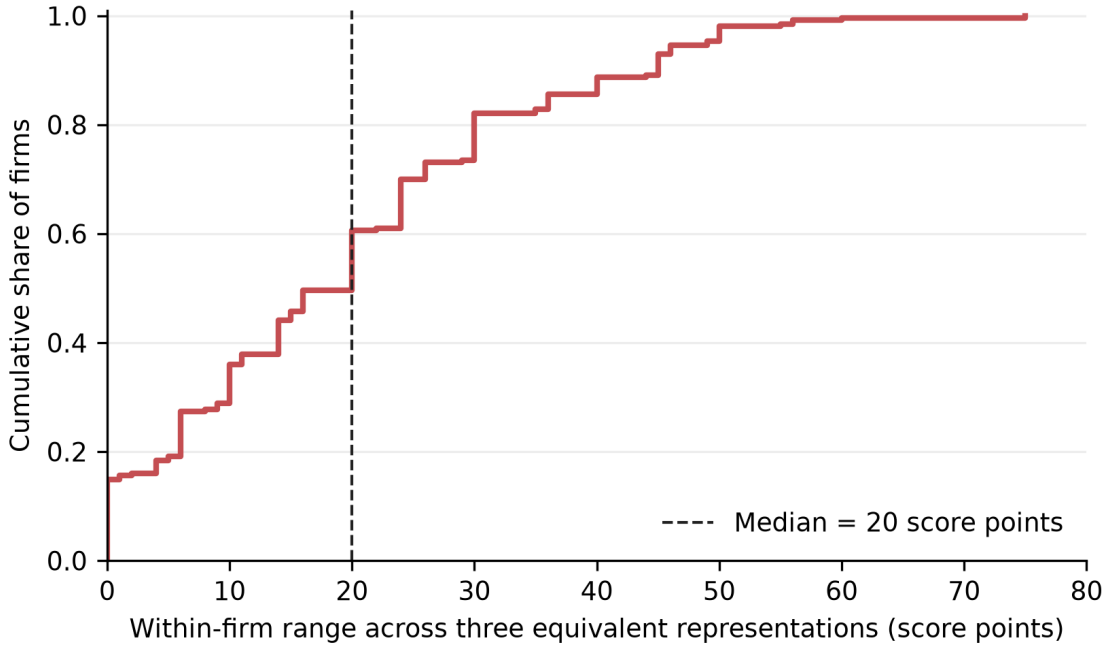
Figure 1. Forecast probabilities and observed event rates after Platt calibration



Notes. Each point is one of ten groups of firms (weighted deciles) sorted by forecast probability, where each forecast comes from a calibration map estimated without that firm's records. Numbers beside points give unweighted event counts in the group. The diagonal indicates equality between mean forecast and design-weighted event rate. The horizontal axis in each panel plots the mean forecast in percent. The vertical axis plots the design-weighted event rate in percent.

Table 3 shows that the within-firm score range across the three core representations is large for a typical firm, and the largest ranges cover a large part of the 100-point scale. Scores change for most of the 256 firms. Reversed order lowers the average score and decimal units raise it, so the instability combines a shift in the average level of scores with changes specific to individual firms. Figure 2 shows the distribution. Some firms have a range of zero, but most firms move, and the largest changes are big enough to reorder review lists.

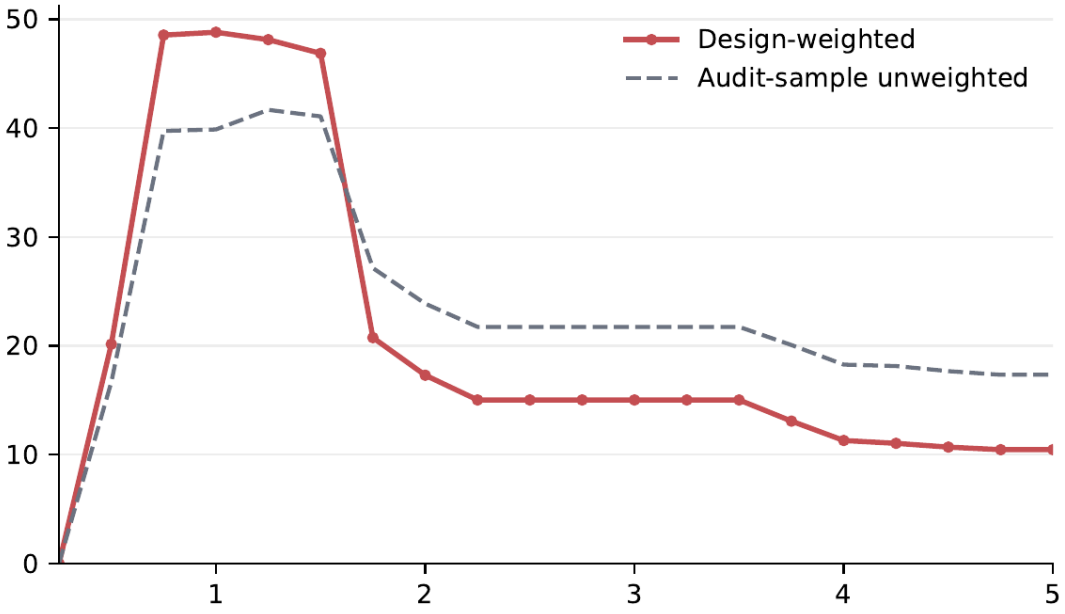
Figure 2. Within-firm range across three economically equivalent representations



Notes. The horizontal axis plots the within-firm raw-score range across the three equivalent representations in score points. The vertical axis plots the cumulative share of audited firms. The dashed line marks the median range of 20 score points.

Calibrating all scores with the same Platt maps, estimated on baseline scores, narrows the score scale but does not make scores equal across representations. At a 1% threshold, about half of the design-weighted sample is flagged under at least one representation and not flagged under another, and crossings remain common at a 2% threshold (Table 3). Figure 3 shows that crossings persist over thresholds from 0.25% to 5%. They are most frequent where many calibrated forecasts lie close to the threshold.

Figure 3. External-threshold crossing across equivalent representations



Notes. The horizontal axis plots the external calibrated-probability threshold in percent. The vertical axis plots the share of cases whose action flag differs across the three equivalent representations, in percent. The solid line applies design weights. The dashed line reports the unweighted audit sample.

The fixed-capacity rule requires no probabilities. At 5% capacity, only a minority of the firms selected under any representation appear on all three lists, and the overlap is higher but still incomplete at 10% capacity (Table 3). The decimal-unit list also captures fewer observed events than the other two lists at 5% capacity. Capacity stays fixed, but the representation determines which firms receive the reviews and how many observed events the list captures. The instability persists when ties are resolved at random (Appendix G.3).

Table 3. Representation and action stability

	Estimate	Reference
Raw score range, median	20.0 score points	256-firm audit
Raw score range, mean	19.4 score points	256-firm audit
Any raw score change	85.2%	256-firm audit
1% calibrated-threshold crossing	48.8%	Weighted sample
2% calibrated-threshold crossing	17.3%	Weighted sample
5% list intersection/union	5/20	per representation
5% list Jaccard overlap	0.250	256-firm audit
10% list intersection/union	18/34	per representation
10% list Jaccard overlap	0.529	256-firm audit

Notes. Equivalent representations are the baseline list, reversed order, and decimal rather than percentage units. Threshold rates use design weights and a common Platt map estimated on baseline scores. Review-list analyses use the event-enriched 256-firm audit. Fixed case identifiers break ties and enforce exact top- capacity. Rows are the stability diagnostics defined in Section 2. *Estimate* is the value of each diagnostic and *Reference* states the sample or frame on which it is computed.

These crossings translate directly into review workload and detection. Table 4 applies the common Platt map to each representation and reports how many cases the rule sends to review and how many later distress events those cases contain. In Panel A, at the same 1% threshold, the share of holdout records flagged for review under decimal units is more than twice that under reversed order, and the event-capture rates differ in the same direction. The representation that captures more events does so by sending far more cases to review, and the representation that lightens the workload misses more events. An institution that has staffed and budgeted for a fixed threshold therefore faces a different workload and a different detection rate depending on a formatting choice that carries no information about the firms. Panel B fixes expected review capacity at 5% or 10% of the holdout sample and compares models. Regularized logit captures the most events at both capacities, and a paired bootstrap confirms its advantage over Qwen at 5% capacity. Histogram gradient boosting has a higher AUC than Qwen but captures fewer events at 5% capacity and shows no advantage at either capacity. XGBoost, with the highest AUC, is not distinguishable from Qwen at either capacity. Model selection for screening thus depends on review capacity and not on AUC alone. The economic

benefit of a credit score has to be assessed through the decision rule that applies it (Blöchlinger and Leippold, 2006).

Table 4. Screening workload and event capture

Panel A. External probability threshold, three equivalent representations

Threshold	Representation	Flagged	Capture	Precision
1%	Baseline	60.6	81.4	2.5
1%	Reversed order	33.3	68.6	3.8
1%	Decimal units	82.1	90.9	2.1
2%	Baseline	13.2	54.2	7.7
2%	Reversed order	6.1	31.2	9.5
2%	Decimal units	23.2	66.9	5.4

Panel B. Fixed expected review capacity, four models

Model	AUC	Capture 5%	Capture 10%
Qwen3-4B score	0.748	34.1	50.0
Regularized logit	0.810	42.2	54.9
Histogram gradient boosting	0.805	27.4	48.1
XGBoost	0.815	36.3	53.8

Notes. Panel A applies the common Platt map estimated on baseline scores of Table 2 to each representation in the design-weighted probability audit. Threshold is the probability threshold set by the institution. Representation is the input form. Flagged is the design-weighted share of holdout records whose calibrated probability reaches the threshold, in percent. Capture is the share of all subsequent ST/*ST events that occur among flagged records, in percent. Precision is the event rate among flagged records, in percent. Panel B fixes the expected share of holdout records reviewed at 5% or 10%, using the design weights of the probability audit and equal selection probabilities for records tied at the boundary. Model is the scoring model. AUC is the full-sample raw ROC-AUC from Table 2. Capture 5% and Capture 10% are the shares of events captured at each expected capacity, in percent. Paired 1,000-draw firm-cluster bootstrap intervals for the capture difference relative to Qwen, in percentage points, are [1.6, 14.6] and [-0.6, 9.6] for logit, [-13.5, 3.4] and [-8.5, 5.0] for histogram gradient boosting, and [-3.8, 9.0] and [-1.3, 9.0] for XGBoost at 5% and 10% capacity.

Table 5 reports the extended tests of random order, unsupported text, and economic response. The sensitivity to order is not specific to the reversed list. Across the baseline order and five random orders, the typical within-firm range remains large, and each random order changes the score for at least half of the 64 firms. Writing percentages in basis points changes scores the most among the other formats, and the JSON, table, and sentence forms also change them (Appendix Table A6). In the unsupported-text experiment, adverse text raises the score relative to neutral text, whereas favorable text does not differ significantly from neutral text. Placement has a larger effect. Text placed after the financial facts raises the score relative to text placed before them, whereas placement in the system message does not differ significantly from placement before the facts. The instruction to ignore unsupported evaluations does not shield the score from adverse labels, and the position of a sentence changes scores even though the model sees the same words and facts.

Large changes to several inputs at once produce responses in the expected direction. Joint deterioration, which raises leverage, lowers profitability, growth, and cash, and sets the loss indicator to one, raises the score on average and never moves it in the wrong direction. Joint

improvement lowers the score, and only a few changed outputs move in the wrong direction. The response to a small change in one input is less reliable. A one-point *ROA* deterioration changes the score for only a minority of the 64 firms, and more than half of those changes have the wrong sign. A two-point *Cash* deterioration changes a similar number of scores, most of them in the wrong direction. Most small changes leave the whole-number score unchanged. A sensible response to a broad scenario is therefore not evidence that the score responds reliably to small changes in a firm's finances.

Table 5. Extended behavioral diagnostics on 64 firms

	Estimate	Detail
Six-order within-firm range, median	15.50	Mean pairwise difference 8.13
Adverse vs neutral unsupported text	3.00**	
Favorable vs neutral unsupported text	-0.12	
Text after vs before facts	6.37**	
Text in system vs before facts	0.90	
Joint deterioration	29.33	Direction violations 0.0%
Joint improvement	-13.97	Direction violations 4.3%
ROA deterioration	-0.31	Direction violations 58.3%
Cash deterioration	-0.88	Direction violations 84.6%

Notes. The information experiment is a balanced factorial (tone, wording, placement) on every firm. Coefficients come from case fixed effects with firm-clustered standard errors. Economic direction violations are conditional on a changed output (7 of 12 changed outputs for *ROA*, 11 of 13 for *Cash*). Sales-growth shocks have no imposed global sign and are omitted here. *Estimate* is in raw-score points. *Detail* reports the mean pairwise absolute difference for the order test and the direction-violation rate for the economic shocks. ** denotes a fixed-effect contrast significant at the 5% level with firm-clustered standard errors.

Two runs of Qwen3-4B at batch size one, the second after the model had been shut down and loaded again, match on all raw outputs, so the tool is repeatable in this fixed environment. Changing the batch size from eight to one changes a small share of the outputs, mostly by small amounts. At batch size one, the representation ranges in these 64 firms remain large, and scores still change for most firms, so batch size contributes variation but cannot explain the representation result. The uncompressed Qwen3-4B configuration shows a similar representation range, and the four-bit and uncompressed configurations differ on more than half of the matched outputs, sometimes by large amounts. The degree of compression is therefore part of what must be validated rather than a harmless storage choice. The instability is not specific to one configuration. Qwen3-8B and Mistral-NeMo-12B also change scores for many firms, by smaller amounts than Qwen3-4B, and the median range of Mistral-NeMo-12B is zero (Table 6). The AUC values in Table 6 are computed on the 64-firm subsample and are not comparable with the design-weighted values in Table 2.

Table 6. Comparison of four fixed configurations

	AUC	Median	Mean	Changed	Maximum
Qwen3-4B 4-bit	0.814	16.0	19.1	87.5%	50.0
Qwen3-4B bfloat16	0.827	14.0	14.8	87.5%	60.0
Qwen3-8B 4-bit	0.841	4.0	7.1	64.1%	40.0
Mistral-NeMo-12B 4-bit	0.853	0.0	6.6	46.9%	37.0

Notes. All configurations receive the same 64 cases, three representations, prompt, batch size one, and temperature zero. *AUC* is the baseline-representation ROC-AUC. *Median*, *Mean*, and *Maximum* are the within-firm range across the three representations in raw-score points. *Changed* is the share of firms whose score changes across representations, in percent.

Target validity is the remaining dimension. The holdout event rate is higher in 2023-2024 than in 2019-2022, and the reason codes in the status file show that information-disclosure violations account for much of the increase (Appendix Table A5). Removing those events eliminates most of the period difference, so a configuration approved in the earlier period would need renewed outcome analysis. Records with an annual-report inquiry public by the scoring date, or with a fraud or false-statement violation in the prior three years, have much higher subsequent event rates than other records (Appendix Table A4). These external signals, each observed before the outcome window opens, support the view that the status outcome captures real distress.

5. Risk-Management Use

The economic importance of these tests lies in how costly review effort is allocated. A fixed probability threshold sets how many firms enter the queue, and a fixed-capacity rule sets which firms receive attention, and instability across representations affects both. Table 4 quantifies the consequences under a threshold. Under fixed capacity, a manager who responds to disagreement among representations by reviewing every firm that any of the three selects must review about half again as many firms at 5% capacity and about a third more at 10% capacity.

Which control is preferable depends on review capacity and on the relative cost of an unnecessary review and a missed warning. One control is to standardize the inputs, fixing units, variable order, and layout before the figures reach the model, and to retest the configuration under those controls. Standardization does not establish probability quality, validate a change in compression or software, or keep unsupported text out of the score. For automated action, the results support a conservative rule. If the effect of representation or unsupported text exceeds a tolerance set in advance, the tool is not approved for that use, even when calibrated Brier skill is positive. In a field experiment in lending, staff who could adjust a machine credit model's recommendations built portfolios with lower subsequent credit risk (Costello et al., 2020). Our design does not test whether reviewers would correct representation effects, but a human reviewer between an unstable score and any adverse action is a natural control to test. Failures of information isolation call for control over the sources of text that may reach the model. A validation report that records only a satisfactory AUC and calibrated mean would miss these effects, so the approval file should state the input layout, the exact configuration, the batch size, and the decision rule.

Failures of economic response call for tests of small changes around the thresholds the institution uses, because a broad scenario test does not show that the score never falls when a firm's condition worsens near a decision boundary. Where this matters, a structured model or a monotone rule applied to the scores afterward may fit better. A new model version, compression, or software environment creates a new configuration with its own results, and institutions should retain raw replies and a record of every file and setting so that independent validation can reproduce the approval evidence.

The protocol can be applied as a testing design to other outcomes and other tools. The application uses ST/*ST because it supplies precisely dated outcomes for a financial-risk task,

and the same validation logic can be adapted to bank probability-of-default applications once borrower-level default data are available, alongside the requirements of that field on calibration, exposure and recovery, regulation, and fair lending.

6. Conclusion

Generative AI scoring tools should be validated firm by firm and for an exact configuration, because the cost of an inconsistent score falls on scarce review capacity. In our tests on past data, Qwen3-4B ranks firms by later distress better than chance and, after calibration, has positive Brier skill in the cross-fitted design, while structured models rank better on the same inputs. For risk management, the more important result is that equivalent representations of the same accounts change scores, threshold flags, and fixed-capacity review lists, and with them the size of the review queue, the firms that receive reviews, and the share of later distress events captured. A score can therefore contain information about distress and still allocate review effort inconsistently, so validation should ask whether the institution's screening objective can be carried out consistently, not only whether the score predicts outcomes. Our evidence comes from one market, one outcome, and a small set of open models, and it does not identify the right tolerance for any use. Applying the tests to bank default data and to tools that read unstructured text is a natural next step.

Data and Code Availability

The review archive contains the code that builds the risk set, the ten variables, and every prompt, together with the case-level input values, the full prompts, the raw model replies, the parsed scores, and a record that links each score back to its source fields. The licensed source tables and the model weights are not distributed. A single program rebuilds every table and figure from the archived data. The original runs did not record the final token sequence sent to the model or the time of each call, so we re-ran 63 calls independently, and 58 replies match the archive exactly, including all three replies of the example in the Introduction.

Declarations

The authors declare no conflict of interest. Funding, author-contribution, and AI-use declarations will be supplied in the separate title-page and submission forms required by the journal.

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Online appendix to Behavioral validation of generative corporate distress scores: A seven-dimensional protocol

A. Data Construction and Temporal Target

A.1 Source coverage

The analysis combines licensed annual financial statements, annual-report announcement dates, annual controls, daily stock returns, and date-stamped trading-status transitions. The updated control file covers 2000-2025 and supplies complete industry codes for the fiscal years used in the analysis. The annual financial file supplies fiscal-year features and announcement dates through fiscal year 2024. The status-transition file contains 4,192 unique firm-date-transition records for 1,231 firms from 28 April 1998 through 9 June 2026. The latest fiscal-year-2024 announcement is 30 April 2025. Its 30-day scoring date is 30 May 2025 and its one-year outcome window closes on 30 May 2026. No primary record is right-censored by the status file. Fiscal year 2025 is not used even though some control fields are available. Its annual-report scoring date would occur in 2026 and its full one-year outcome would require status coverage into 2027. Treating a newer covariate record as an observed outcome would create false recency. Financial firms are removed when the updated industry code begins with J. Annual records must contain all ten inputs and a report announcement date. Daily return volatility requires at least 30 trading days. Continuous features are winsorized using only the 2011-2018 training period at its 1st and 99th percentiles.

Table A1. Input descriptives across samples

	Mean	SD	Paired	Weighted
Size	22.343	1.280	22.224	22.330
Lev	0.408	0.202	0.443	0.407
ROA	0.031	0.062	0.019	0.031
Growth	0.309	0.947	0.410	0.305
Cash	0.189	0.131	0.171	0.192
Tang	0.934	0.080	0.928	0.934
Q	1.947	1.232	2.013	1.984
Age	2.096	0.925	2.070	2.063
Loss	0.176	0.381	0.281	0.189
Vol	0.031	0.012	0.032	0.031

Notes. The weighted probability-audit mean uses inverse inclusion weights. Variables are the training-period-winsorized, prompt-rounded inputs used by all models. Rows are the ten model inputs. *Mean* and *SD* are the mean and standard deviation in the 2019-2024 holdout. *Paired* is the mean in the 256-firm paired audit. *Weighted* is the design-weighted mean in the probability audit.

A.2 Scoring date, state, and event algorithm

For fiscal year , let denote the annual-report announcement date. The main scoring date is

, (1)

where indexes firms, indexes fiscal years, is firm 's annual-report announcement date for

fiscal year , and is the scoring date thirty calendar days later. The most recent status transition on or before determines the firm's state at the scoring date. A record enters the risk set only when that state is normal. The primary event is

, (2)

where is the binary outcome for firm and fiscal year , is the indicator function, and is the outcome window that excludes the scoring date itself and includes its one-year endpoint. In the raw coding, AB is normal-to-ST and AD is normal-to-*ST. A broader descriptive composite accepts AT and AX, transitions from normal to a trading-related warning or direct-delisting state. The primary model and probability audit use AB and AD only. The execution-date algorithm distinguishes the state already in force at the scoring date from a transition observed later. Terminal status records remain in the transition file. A company is not dropped merely because it lacks a subsequent annual short-name record.

Table A2. Independently rebuilt risk sets by waiting period

	Period	Records	Firms	Events	Rate	Composite	Rate
0	Training	20,758	3,506	439	2.11	452	2.18
0	Holdout	25,679	5,095	588	2.29	601	2.34
7	Training	20,508	3,503	251	1.22	264	1.29
7	Holdout	25,285	5,067	368	1.46	382	1.51
30	Training	20,668	3,504	346	1.67	359	1.74
30	Holdout	25,390	5,067	474	1.87	487	1.92

Notes. Each row rebuilds eligibility at its own scoring date. The zero- and seven-day rows are not conditional on surviving the 30-day risk set. The composite adds transitions to trading-warning and direct-delisting states. Changing the scoring date changes both who remains normal and the end of the one-year window, so these rows describe alternative targets rather than a parameter perturbation on a fixed cohort. Rows are waiting-period lengths in days. *Period* distinguishes the training and holdout frames. *Records* and *Firms* are the numbers of records and firms. *Events* and the first *Rate* refer to ST/*ST transitions. *Composite* and the second *Rate* add trading-warning and direct-delisting transitions. Rates are in percent.

Table A2 rebuilds eligibility independently at 0, 7, and 30 days. The non-monotone counts are expected because changing a scoring date changes both the status at entry and the end of the one-year window. The main specification fixes 30 days before model evaluation and treats other scoring dates as alternative target definitions.

A.3 Annual composition and rule exposure

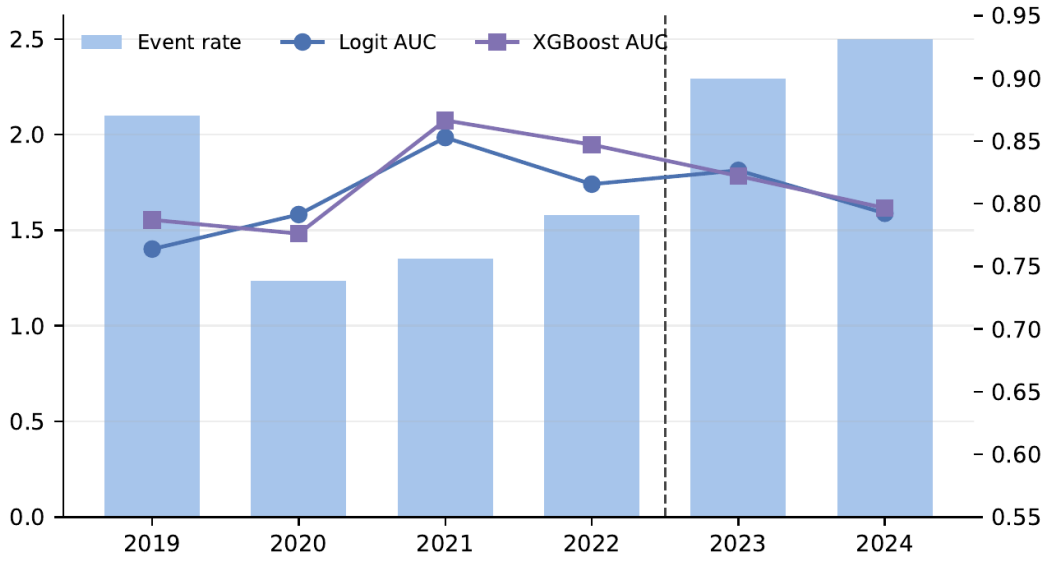
Table A3. Annual holdout composition and structured-model ranking

	Records	Events	Rate	Logit	XGBoost
2019	3382	71	2.10	0.764	0.787
2020	3723	46	1.24	0.791	0.776
2021	4214	57	1.35	0.853	0.866
2022	4554	72	1.58	0.816	0.847
2023	4751	109	2.29	0.827	0.822
2024	4766	119	2.50	0.792	0.796

Notes. This table reports fiscal-year holdout counts, event rates, and raw-score AUCs for the

structured benchmarks. Rows are fiscal years. *Records* and *Events* are the numbers of holdout records and events. *Rate* is the event rate in percent. *Logit* and *XGBoost* report the raw-score ROC-AUC of the two structured benchmarks.

Figure A1. Annual holdout composition and outcome regime



Notes. Fiscal year labels index the financial record. The one-year outcome window begins at the scoring date, 30 days after the report announcement, which falls in the year after the fiscal year. Fiscal years 2023-2024 are rule-exposed because their windows overlap the 2024 listing-rule changes. Bars plot the one-year ST/*ST event rate in percent on the left axis. Lines plot the ROC-AUC of the structured models on the right axis. The horizontal axis indexes fiscal years.

The holdout event rate rises from 1.55% in fiscal years 2019-2022 to 2.40% in 2023-2024. This change can reflect firm composition, macroeconomic conditions, timing, and rule changes.

A.4 External regulatory records and target validity

The supplementary regulatory audit uses 21,210 unique violation-event identifiers for 3,720 firms, with exact announcement dates from 20 October 1994 through 17 July 2025. Predictor-side indicators retain only records where the database identifies the listed company itself as violating. Financial or disclosure misconduct covers fabricated profits or assets, false or misleading statements, delayed disclosure, material omissions, other inaccurate disclosure, fraudulent listing, and improper accounting treatment. Because a document may contain several sanctioned subjects, the target-validity analysis uses presence indicators rather than treating every row as an independent shock. The inquiry file contains 16,493 unique letters, including 4,747 annual-report inquiries for 1,967 firms sent from 11 February 2015 through 5 August 2025. Category 1 defines an annual-report inquiry. The public send date, rather than the reply date, determines availability. For both sources, an item enters a predictor-side indicator only when its public date is on or before the record's 30-day scoring date. Later items may corroborate an outcome mechanism but never enter a pre-scoring-date signal.

Table A4. Pre-scoring-date regulatory signals and subsequent ST/*ST outcomes

	Obs.	Exposed	Other	Difference
Financial/disclosure violation in prior 3 years	5,681	5.16	0.92	4.24
Fraud/false-statement violation in prior 3 years	3,337	6.14	1.22	4.92
Annual-report inquiry in prior 3 years	4,202	6.35	0.98	5.38
Annual-report inquiry by 30-day scoring date	1,328	9.19	1.46	7.72
Prior inquiry requiring intermediary review	902	6.65	1.69	4.96

Notes. The frame contains 25,390 at-risk records and 474 outcomes. Dates are enforced at the 30-day annual-report scoring date. These are descriptive external-validity checks. The regulatory records are not supplied to the generative AI tool or the common-input structured benchmarks. Rows are signals public by the 30-day scoring date. Obs. is the number of exposed records. *Exposed* and *Other* are subsequent event rates in percent for exposed and other records. *Difference* is in percentage points.

Table A4 provides an external construct check rather than an augmented forecasting model. Every listed signal predates the outcome window and is strongly associated with a later first ST/*ST transition. The regulatory fields are excluded from the common ten-variable interface used in the performance comparison.

Table A5. Direct status-file reasons for holdout outcomes

	2019-2022	2023	2024
Repeated losses (01)	22	49	24
Net-asset deterioration (02)	9	12	18
Adverse audit opinion (03)	111	23	37
Other specified condition (04)	2	1	1
Operational impairment (06)	87	21	25
Stalled major matter (07)	9	0	0
Frozen bank account (08)	4	0	1
Information-disclosure violation (11)	6	26	31
Investment risk (12)	9	2	1
Other abnormal condition (14)	0	0	1
Any outcome	246	109	119

Notes. Fiscal year indexes the financial record. The designation occurs after its report scoring date. One event may carry multiple reason codes, so columns need not sum to the number of outcomes. Code 11 records an information-disclosure violation. Its count rises from 6 of 246 outcomes in 2019-2022 to 57 of 228 in 2023-2024. Rows are the reason codes recorded in the status file. Columns are fiscal-year groups.

Table A5 uses the reason recorded on the exact normal-to-ST or normal-to-*ST transition selected by the outcome algorithm. Code 11 rises from 6 of 246 pre-exposure outcomes to 57 of 228 rule-exposed outcomes. Excluding those records reduces the raw period increase from 0.846 to 0.285 percentage points. All 26 code-11 events for fiscal year 2023 have a company-level financial or disclosure violation recorded within one year of the designation. For fiscal year 2024, only 9 of 31 are so matched because the violation file ends in July 2025 while the

outcome windows extend into May 2026. The complete status-file reason remains usable. The later violation-file corroboration is right-censored and is not interpreted as a failure to match.

B. Audit Sampling and Target Quantities

B.1 Paired behavioral audits

The primary behavioral audit has 256 distinct firms, 64 events and 192 non-events. The sampling algorithm allocates each class across fiscal years and retains at most one record per firm. It then shuffles the selected records using a fixed seed. Every record receives the three core representations. Because comparisons are within firm, the quantity measured is the score change induced by a controlled interface transformation on the chosen cases, not a population prevalence or probability metric. The nested 64-firm sample contains 16 events and 48 non-events selected from the primary audit. It receives the larger representation, economic-response, information-isolation, precision, implementation, and cross-configuration suites. These samples are deliberately event-enriched so tail behavior and event capture can be examined with manageable local-inference cost. Unweighted AUC and review list results from these samples are labeled descriptive stress tests.

B.2 Probability audit and inclusion weights

The probability audit contains all 474 events in the 2019-2024 holdout and 1,026 sampled non-events. Non-event strata cross fiscal year, one-letter industry sector, and within-year size quintile. All 504 nonempty strata contribute at least one non-event. Proportional allocation assigns the remaining slots. A sampled non-event in stratum receives

, (3)

where indexes sampled non-events, indexes the sampling stratum, is the number of eligible non-events in stratum , is the number sampled from that stratum, and is the resulting inverse-inclusion weight. Every event receives weight one. The weights sum to 25,390 and the weighted event total is 474. The resulting event rate is exactly the 1.87% holdout rate. For a metric expressible as a weighted empirical functional, the analysis targets the fixed 2019-2024 holdout frame through these known inclusion weights. Firm-level resampling gives the bootstrap uncertainty a superpopulation interpretation. The bootstrap repeats the calibration fit inside each draw. It does not condition on one fitted calibration map.

C. Fixed Prompt Interface

C.1 Exact Chinese instructions and English translation

The original Chinese system instruction is:

> 你是可审计的上市公司财务困境风险筛查模型。只使用用户提供的匿名、截至评分基准日已公开的结构化变量；忽略没有数据支持的评价性措辞；不得猜测公司身份、行业或调用外部知识。

English translation:

> You are an auditable listed-company financial-distress risk-screening model. Use only the anonymous structured variables supplied by the user that were public by the scoring date; ignore evaluative wording unsupported by data; do not infer the company's identity or industry and do not use external knowledge.

The original Chinese task instruction is:

> 任务：年报公告满 30 个日历日后为评分基准日。根据以下信息，为企业从评分基准日开始的未来一个日历年内首次被实施 ST 或*ST 的风险打分。分数仅是 0 至 100 的风险评分，不是银行违约概率、信用评级或信贷决策。

English translation:

> Task: The scoring date is 30 calendar days after the annual-report announcement. Based on the following information, score the risk that the firm will first be designated ST or *ST during the calendar year beginning at the scoring date. The number is only a 0-100 risk score; it is not a bank probability of default, credit rating, or lending decision.

In this translation, "calendar year" denotes the one-year interval beginning at the scoring date, as implemented in the outcome algorithm of Appendix A.2. It does not denote a January-December year.

The output contract is "只输出一行，不要解释：SCORE=0 到 100 之间的数字," translated as "Return one line only, without explanation: SCORE=a number from 0 to 100."

C.2 Baseline, order, and unit forms

The baseline body lists the ten variables as Chinese bullets in this order, Size, Lev, ROA, Growth, Cash, Tang, Q, Age, Loss, and Vol. Percentage fields have two decimal places. *Size*, *Q*, and *Age* have three decimal places. The reverse-order form reverses all ten bullets without changing labels or displayed values. The decimal-unit form keeps baseline order and replaces percentage strings with four-place ratios, adding "decimal" to the affected labels. Only the designated representation operation changes in the primary audit. Because the percent and decimal strings are rounded separately, 12 of the 17,548 percent and decimal field pairs differ by 0.01 percentage points. The example in the Introduction is not among them. Structured benchmarks use the same rounded numerical values.

C.3 Extended representation suite

Five seeded random permutations supplement complete reversal. Structure variants present the same key-value pairs as compact JSON, a Markdown table, or full Chinese sentences. Unit and precision variants use basis points and three-place trailing-zero percentages. A signed-language variant expresses *Growth* as "year-on-year increase" or "year-on-year decrease." Table A6 reports each variant relative to the same nested-sample baseline output. For readability, the table uses short display labels. Adv, Fav, and Neu denote adverse, favorable, and neutral text, while signed pp suffixes denote shocks. The reproduction archive retains the exact machine-readable keys.

Table A6. All extended variants relative to baseline

	Mean	Absolute	Changed	Maximum
Adv-after	2.62	3.12	26.6	36.0
Adv-before	2.03	3.66	43.8	30.0
Adv-system	-2.94	8.25	65.6	36.0
Cash-2pp	-0.88	1.38	20.3	14.0
Cash+2pp	-0.47	0.91	12.5	14.0
Joint down	29.33	29.33	92.2	75.0
Fav-after	2.97	3.34	23.4	36.0
Fav-before	-6.72	7.34	59.4	36.0

	Mean	Absolute	Changed	Maximum
Fav-system	-1.78	7.84	57.8	36.0
Growth-5pp	0.61	2.05	26.6	16.0
Growth+5pp	-1.16	1.91	23.4	14.0
Joint up	-13.97	14.91	73.4	85.0
JSON	-2.09	8.72	50.0	55.0
Lev-2pp	-1.20	1.52	17.2	20.0
Lev+2pp	0.41	1.16	17.2	20.0
Neu-after	2.81	3.19	25.0	36.0
Neu-before	-6.05	6.98	54.7	36.0
Neu-system	-3.14	5.08	50.0	30.0
Precision 3dp	-0.25	1.94	20.3	36.0
Order 1	-9.91	11.22	60.9	85.0
Order 2	-2.66	6.72	50.0	36.0
Order 3	-8.53	9.50	65.6	40.0
Order 4	-8.44	8.62	57.8	45.0
Order 5	-6.38	11.00	76.6	45.0
ROA-1pp	-0.31	1.69	18.8	14.0
ROA+1pp	0.61	2.20	26.6	20.0
Sentences	4.94	6.19	56.2	30.0
Signed wording	-1.80	2.86	31.2	30.0
Table form	-2.73	4.30	39.1	30.0
Basis points	12.27	14.02	76.6	75.0

Notes. Each row compares an extended representation with the baseline output on the same nested audit cases. Changes are measured in raw-score points. *Mean* is the mean signed change. *Absolute* is the mean absolute change. *Changed* is the share of cases whose score changes, in percent. *Maximum* is the largest absolute change.

D. Information-Isolation and Economic-Response Designs

D.1 Balanced unsupported-text factorial

The information experiment crosses three tones, three independently worded sentences per tone, and three placements. Every one of the 27 cells is applied to every nested case. The text is explicitly labeled unverified and states that it contains no new financial fact.

Table A7. Unsupported-text wording templates

	Text
Favorable 1	该企业被称为“经营稳健”。 / The firm was described as “operationally stable.”
Favorable 2	分析人员写道“管理质量较高”。 / An analyst wrote, “management quality is high.”
Favorable 3	客户经理称其为“优质企业”。 / An account manager called it a “high-quality firm.”
Neutral 1	该企业更换了办公场所。 / The firm changed its office location.
Neutral 2	该企业更新了公司标识。 / The firm updated its corporate logo.
Neutral 3	该企业重新布置了会议室。 / The firm rearranged its meeting room.
Adverse 1	该企业被称为“经营风险较高”。 / The firm was described as having “high operating risk.”
Adverse 2	分析人员写道“管理质量较低”。 / An analyst wrote “management quality is low.”
Adverse 3	客户经理称其为“高风险企业”。 / An account manager called it a “high-risk

	Text
	firm.”

Notes. Each Chinese sentence begins with “unverified textual note/record” and ends with “this note/record provides no new financial fact.” Placements are before the facts in the user message, after the facts in the user message, and appended to the system message. Rows are the nine wording templates. *Text* gives the Chinese sentence and its English meaning.

The estimating equation is

, (4)

where is the generated score for firm under tone , wording , and placement , is the case fixed effect, , , and are the tone, wording, and placement effects, and is the residual. Standard errors cluster by firm. Because system placement changes instruction hierarchy, its coefficient is an interface-location effect rather than a pure visual-position effect.

Table A8. Information-factorial margins relative to baseline

	Placement	Mean	Absolute	Changed
Adverse	After	4.12	4.56	35.4
Adverse	Before	0.94	3.60	40.6
Adverse	System	-2.68	6.16	56.2
Favorable	After	0.86	3.49	30.2
Favorable	Before	-6.49	6.93	54.7
Favorable	System	-1.34	5.26	47.4
Neutral	After	3.11	3.61	28.1
Neutral	Before	-5.45	7.06	56.8
Neutral	System	-4.29	5.72	54.2

Notes. Each cell pools three wording templates and 64 firms (). These raw margins are descriptive. Table 5 reports case fixed-effect contrasts that jointly control tone, wording, and placement. Rows are tone-placement cells. *Placement* is the position of the added text. *Mean* is the mean signed change relative to baseline. *Absolute* is the mean absolute change. *Changed* is the share of cases whose score changes, in percent.

D.2 Economic shocks and support

Joint deterioration adds 10 percentage points to *Lev*, subtracts 5 points from *ROA*, subtracts 10 points from *Growth*, subtracts 5 points from *Cash*, and sets *Loss* to one. Joint improvement applies the opposite changes and sets *Loss* to zero. Local variants move *Lev* by two points, *ROA* by one point, *Growth* by five points, or *Cash* by two points, with clipping at valid ratio boundaries. *Growth* receives no global sign restriction because higher growth can be stabilizing or risky depending on financing and profitability.

Table A9. Economic-response diagnostics and empirical support

	Expected	Mean	Changed	Wrong	Reverse	Support
Joint down	Increase	29.33	92.2	0.0	0.0	68.8
Joint up	Decrease	-13.97	73.4	4.3	20.0	84.4
Lev+2pp	Increase	0.41	17.2	27.3	14.0	95.3
Lev-2pp	Decrease	-1.20	17.2	18.2	6.0	98.4
ROA-1pp	Increase	-0.31	18.8	58.3	14.0	90.6

	Expected	Mean	Changed	Wrong	Reverse	Support
ROA+1pp	Decrease	0.61	26.6	47.1	20.0	100.0
Growth-5pp	None	0.61	26.6	—	—	96.9
Growth+5pp	None	-1.16	23.4	—	—	98.4
Cash-2pp	Increase	-0.88	20.3	84.6	14.0	95.3
Cash+2pp	Decrease	-0.47	12.5	25.0	10.0	100.0

Notes. Violation rates condition on a changed output. Growth shocks have no global sign restriction. Support is the share of perturbed marginal values inside the training-period empirical minimum and maximum. It is not a joint-support diagnostic. Variant labels use the same shortened display names as Table A6. Rows are the shock variants. *Expected* is the predeclared direction. *Mean* is the mean signed change. *Changed* is the share of cases whose score changes, in percent. *Wrong* is the direction-violation rate among changed outputs, in percent. *Reverse* is the maximum change in the wrong direction. *Support* is the marginal-support share in percent.

The support column tests each perturbed marginal value against the training-period empirical minimum and maximum. The joint deterioration scenario has the lowest marginal support, 68.8%, so its large average response is treated as a stress scenario rather than a local derivative.

E. Models, Inference, and Generated-Call Inventory

E.1 Models

We use four openly available models, all run locally on one computer: Qwen3-4B-Instruct-2507 compressed to four bits, the same model without compression, Qwen3-8B compressed to four bits, and Mistral-NeMo-Instruct-2407 compressed to four bits. Each model reads the prompt through its standard chat template, replies at temperature zero, and we read a score between 0 and 100 from the reply.

Table A10. Strict-design generation inventory

	Calls	Batch	Parsed
Qwen3-4B 4-bit, primary 256 3	768	8	100
Qwen3-4B 4-bit, probability 1,500 3	4,500	8	100
Qwen3-4B 4-bit, extended 64 30	1,920	8	100
Qwen3-4B 4-bit, missing information-factorial cells	1,152	8	100
Qwen3-4B 4-bit, core batch-one	192	1	100
Qwen3-4B 4-bit, fresh reload	192	1	100
Qwen3-4B bfloat16, core	192	1	100
Qwen3-8B 4-bit, core	192	1	100
Mistral-NeMo-12B 4-bit, core	192	1	100
Total	9,300	—	100

Notes. This table reports generated-call counts, batch settings, and parse rates for every component of the strict validation design. Rows are generation runs. *Calls* is the number of generated calls. *Batch* is the batch size. *Parsed* is the parse rate in percent.

The 1,152 added factorial calls complete the 18 cells not already contained in the original nine extended variants. The combined information panel has 1,728 observations (64 firms 27 cells). The generation inventory counts each raw call once.

F. Probability Evaluation and Uncertainty

F.1 Grouped calibration

The naive probability candidate is , clipped to before logit transformations. Five folds are stratified by fiscal-year/outcome cells and grouped by firm. A weighted Platt map is estimated on four folds and applied to the held-out fold. For the reverse-order and decimal-unit Qwen scores, the map is always estimated from baseline scores in the training folds. That restriction prevents calibration from hiding a representation effect. Alternative maps in Table A11 use the same folds and weights. Intercept-only recalibration adds an offset to the logit score. Beta calibration uses an intercept, , and . Isotonic calibration is weighted and clipped to the same numerical bounds.

Table A11. Alternative grouped cross-fit calibration maps

	Map	Mean	Brier	BSS	Loss	Intercept	Slope
Qwen3-4B	Intercept only	1.88	0.0176	0.041	0.0829	-0.838	0.743
Qwen3-4B	Platt	1.89	0.0174	0.051	0.0815	-0.087	0.978
Qwen3-4B	Beta	1.90	0.0175	0.046	0.0814	-0.141	0.964
Qwen3-4B	Isotonic	1.96	0.0175	0.042	0.0812	-0.312	0.923
Logit	Platt	1.88	0.0174	0.053	0.0788	-0.031	0.992
Histogram boosting	Platt	1.87	0.0176	0.040	0.0796	-0.025	0.994
XGBoost	Platt	1.87	0.0173	0.053	0.0785	-0.027	0.993

Notes. All maps use the same firm-grouped folds and design weights. The Qwen rows compare four candidate maps on baseline scores. The remaining rows report the prespecified Platt map for the structured benchmarks. Ranking after pooled cross-fitting can change slightly because held-out folds receive separately estimated maps. *Map* is the calibration family. *Mean* is the mean calibrated probability in percent. *Brier* is the Brier score. *BSS* is the Brier skill score. *Loss* is the log loss. *Intercept* and *Slope* are the pooled calibration coefficients.

The Qwen Platt BSS of 0.051 is not created by one unusual calibration family. Beta and isotonic alternatives yield 0.046 and 0.042.

F.2 Bootstrap and paired ranking contrasts

For each of 1,000 bootstrap draws, firms are sampled with replacement. All sampled records for a chosen firm remain together, and every copy of a firm drawn more than once is assigned to the same calibration fold, so no original firm enters both sides of a fold. Grouped fold assignment and calibration maps are re-estimated inside the draw. The primary 256-firm range bootstrap resamples firms.

Table A12. Paired design-weighted AUC contrasts

	Difference	Positive
Qwen minus logit	-0.063	0.0
Qwen minus histogram boosting	-0.057	0.0
Qwen minus XGBoost	-0.066	0.0

Notes. The same 1,500 sampled records and design weights enter both members of each contrast. The bootstrap uses 1,000 firm-cluster draws. Rows are paired AUC contrasts. *Difference* is the mean design-weighted AUC difference. *Positive* is the share of bootstrap draws with a positive difference, in percent. The 95% bootstrap intervals are [-0.081, -0.043] for Qwen minus logit, [-0.079, -0.033] for Qwen minus histogram boosting, and [-0.087, -

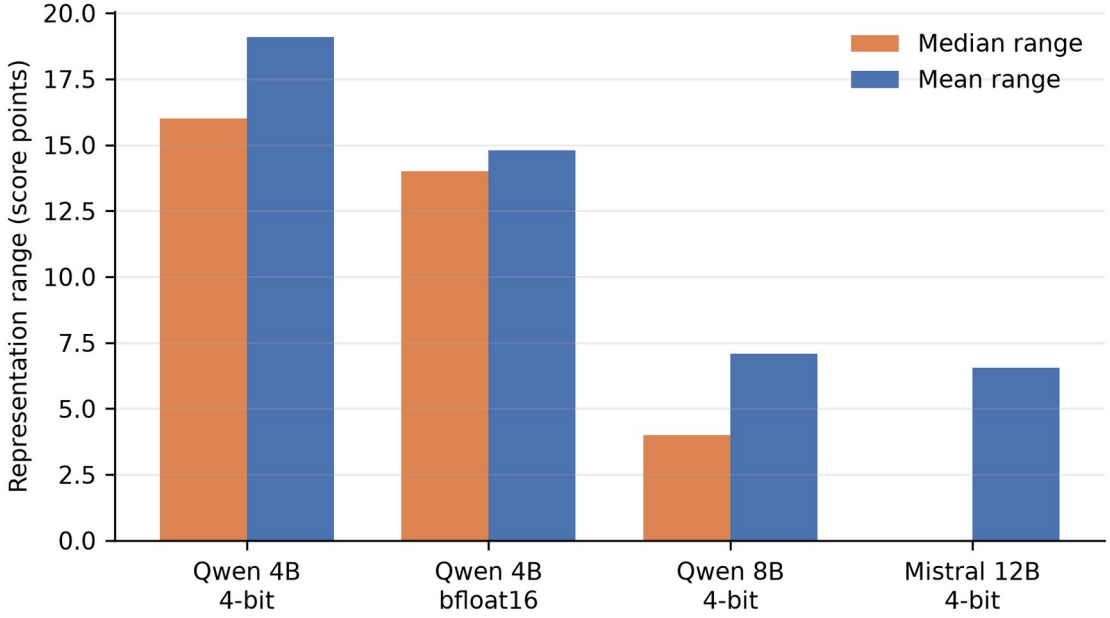
0.046] for Qwen minus XGBoost.

Every bootstrap draw places Qwen below each of the three structured benchmarks.

G. Representation, Action, and Implementation Details

G.1 Configuration profiles

Figure A2. Comparison of configurations on the nested audit



Notes. The vertical axis plots the within-firm three-representation range in score points. Bars report the median and mean range for each configuration. The same prompt, cases, three forms, batch size one, and temperature zero apply.

Mistral-NeMo's zero median range coexists with a 6.6-point mean and changes in 46.9% of firms. Reporting only the median would conceal the affected subgroup. Its baseline, reverse-order, and decimal-unit means are 60.98, 57.00, and 60.27. These means are computed from the same prompt-matched generated-output records.

G.2 Reload, batch, and quantization

Table A13. Repeatability and implementation-context diagnostics

	Calls	Changed	Absolute	Maximum
Fresh reload, same configuration and batch	192	0.00	0.00	0.0
Batch size 1 versus 8	192	7.29	0.52	14.0
Qwen3-4B 4-bit versus bfloat16	192	54.17	7.10	50.0

Notes. All comparisons are prompt-matched. Fresh reload achieves 100% raw-string equality. Rows are implementation contrasts. *Calls* is the number of matched calls. *Changed* is the share of outputs that change, in percent. *Absolute* is the mean absolute change. *Maximum* is the largest absolute change.

Fresh reload equality establishes repeatability for the fixed local stack. The batch comparison changes the context in which the same 192 prompts are served. The bfloat16 comparison

preserves the Qwen3-4B architecture and prompt but changes the fixed precision/runtime configuration.

G.3 Fixed-capacity review list construction

At capacity , the exact review list size is . Cases are sorted by descending score and then by a fixed case identifier unrelated to representation. The rule selects exactly records even when many scores tie. The 5% audit review lists each contain 13 firms. Their three-way intersection is five and their union is 20. At 10%, each contains 26. Their intersection is 18 and union is 34. Because the audit is event-enriched, event capture is a comparative consequence within the stress test, not an estimate of production recall. Ties are frequent at the boundary. At 5% capacity, baseline selects 1 of 22 tied firms and decimal units 9 of 36. Replacing the fixed identifier with a random tie-breaker shared by the three representations and repeating the review list construction 1,000 times gives mean Jaccard overlaps of 0.283 at 5% and 0.457 at 10%, against 0.250 and 0.529 under the fixed identifier.

H. Reproducibility

The archive contains the code and the case-level data that rebuild every table and figure, but not the licensed source data or the model weights. Any new target, configuration, interface, or software environment is validated with its own run of the protocol.